

AURORA-12

A fully synthetic benchmark for evidence-linked poster generation

PostEx Synthetic Research Collective

Fictional Methods Laboratory · Example Research Campus · no real authors or institutions

POSTEX-SYNTH-2026-001 · fully synthetic · CC0-1.0 · no DOI

Fictional lab logo

1,240
synthetic
profiles

12
anonymous
model features

4
deterministic
cohort
simulations

Communication test

Can a realistic multi-cohort manuscript exercise evidence-linked poster generation without redistributing a real paper or implying biomedical evidence?

Evidence: Abstract and Introduction, pp. 1-2

Deterministic workflow

- Fixed seed generated 1,240 profiles
- 480 anonymous transcript-like features
- Nested elastic-net selection
- 12-feature score and shift testing

FOUR SYNTHETIC COHORTS

Four deterministic synthetic cohorts

- Discovery n=620 · Internal validation n=310
- External A n=186 · External B n=124
- No participants, specimens or external datasets

Evidence: Abstract and Methods, pp. 1-2

Generation to validation

Synthetic benchmark workflow

All profiles, features and outcomes are authored simulation fixtures

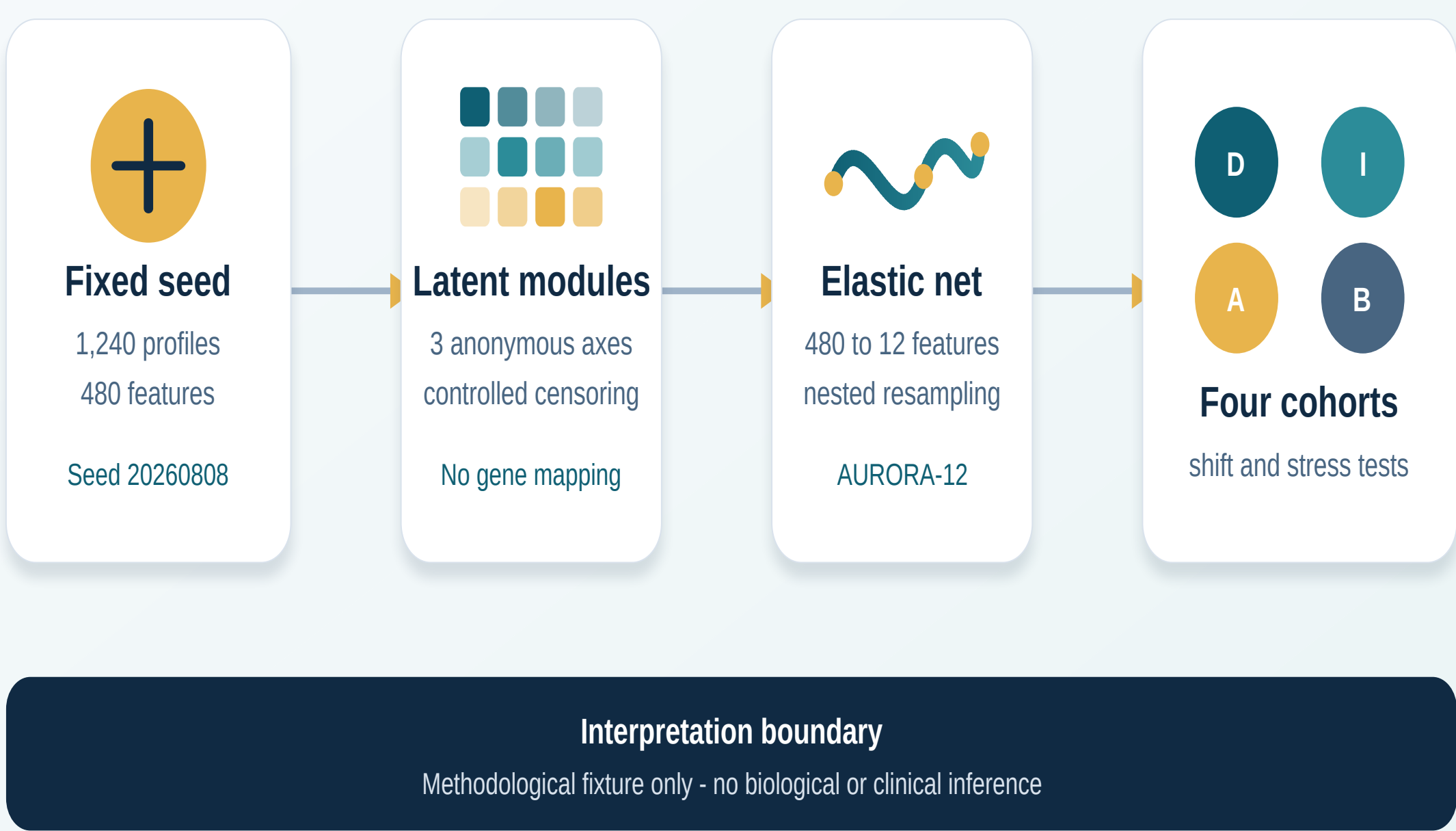


Figure 1 · Deterministic generation and validation workflow
Authored synthetic fixture · manuscript p. 2

External C-indices remained 0.75 and 0.74

Synthetic baseline: 0.66 and 0.65 · software fixture, not clinical validation

Evidence: Results, pp. 1 and 3; Figure 2

Discrimination across shifts

Authored discrimination benchmark

Concordance index - synthetic values only



Figure 2 · Authored discrimination values across cohort shifts
Authored synthetic fixture · manuscript p. 3

Calibration and stress boundary

- External calibration slopes were 0.97 and 0.95.
- External absolute errors were 0.031 and 0.038.
- Aggregate coefficient-sign stability was 92% across 500 resamples.
- The most severe pre-specified shift reduced C-index by 0.04.

Evidence: Results, pp. 1, 3-4; Figures 3-4

External validation retained discrimination: C-index 0.75 / 0.74

Calibration & stability

92%

coefficient-sign stability

0.97 / 0.95

external calibration slopes

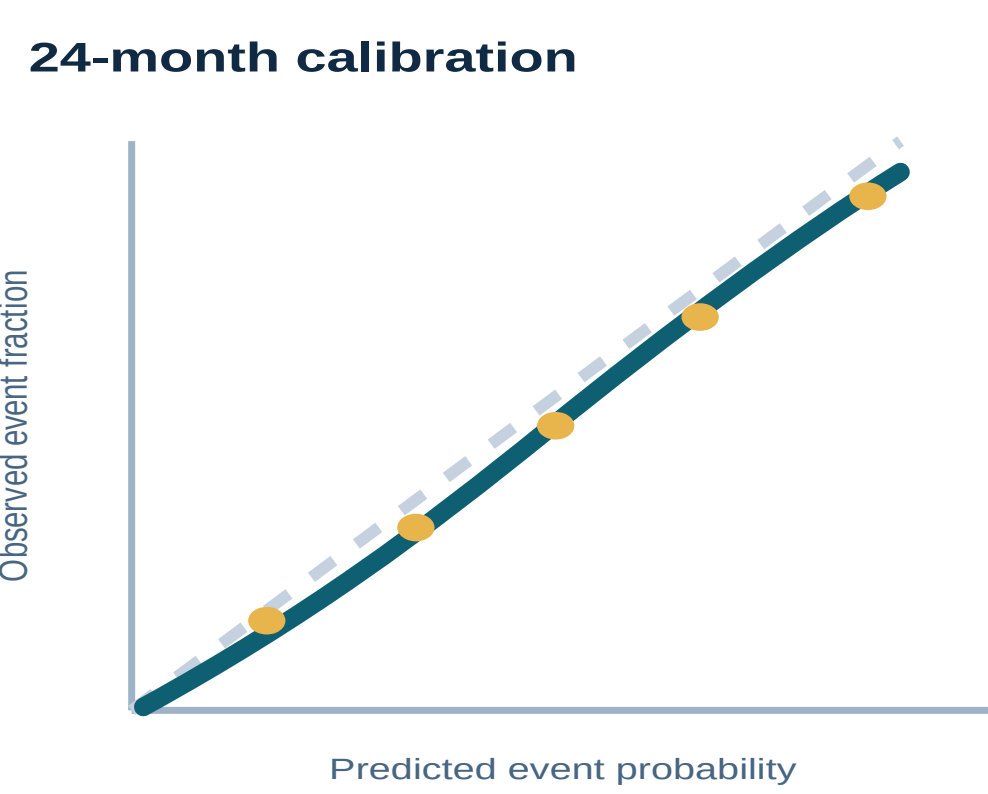
0.031 / 0.038

external absolute errors

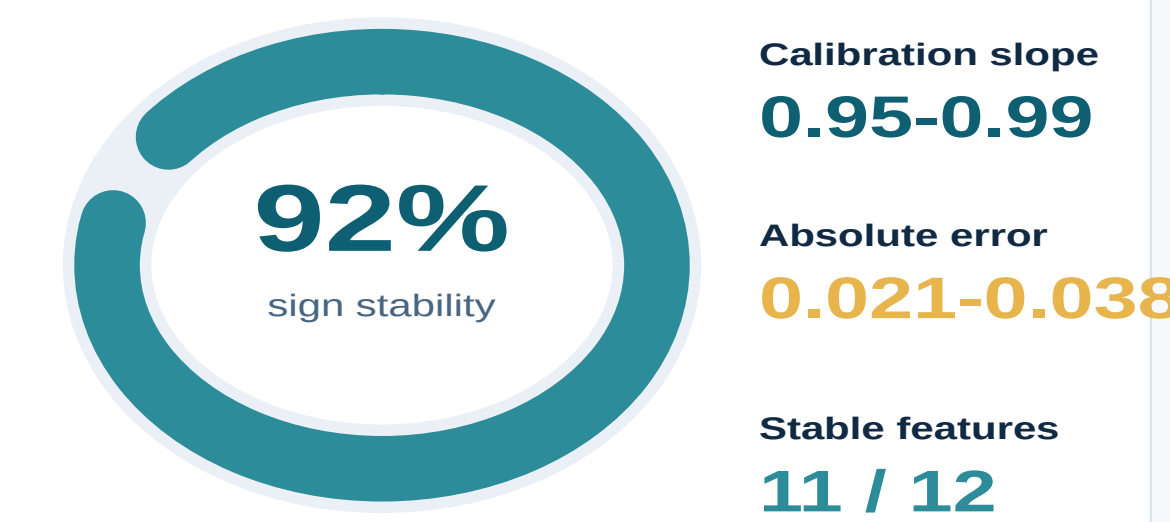
Evidence: Stability and calibration, p. 4

Calibration and coefficient stability

Authored evaluation values - no patient-level data



500 bootstrap resamples



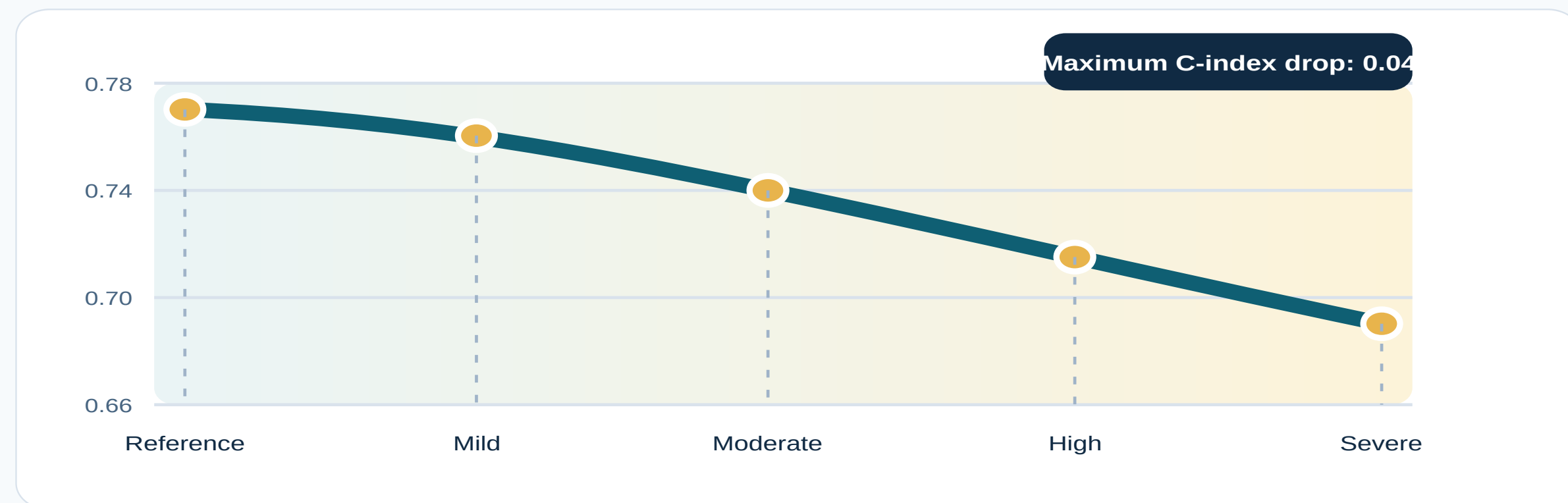
Synthetic stability supports fixture QA only

Figure 3 · Calibration and coefficient-sign stability
Authored synthetic fixture · manuscript p. 4

Pre-specified stress test

Pre-specified stress-test boundary

Increasing batch shift and latent-module imbalance



Do not interpret this authored curve as biological transportability or clinical robustness

Figure 4 · Stress-test boundary under increasing shift
Authored synthetic fixture · manuscript p. 4

Supported conclusion

A rights-safe methodological test fixture

- No biological, prognostic or clinical inference
- All cohorts, values and identities are fictional

Evidence: Discussion and Conclusion, p. 5